

COVID-19 Data Analytics Project

Missing Value Analysis & Handling

Tool: Python (Pandas, Matplotlib, Seaborn), SQL

Dataset: Covid19_dataset.csv (187 countries, 15 columns)

Dashboard: Power BI / Tableau

1. Project Overview

- Understand the dataset structure and columns
- Introduce random missing values to simulate real-world dirty data
- Analyse missing values using Python
- Apply different missing value handling strategies
- Visualise results in a professional dashboard
- Export a clean dataset ready for Power BI or Tableau

2. Dataset Summary

- Rows: 187
- Columns: 15
- Key Features: Case counts (Confirmed, Deaths, Recovered, Active, New cases, New deaths)
- Rate metrics (Deaths / 100 Cases, Recovered / 100 Cases, Deaths / 100 Recovered)
- Weekly trend (Confirmed last week, 1 week change, 1 week % increase)
- Region classification (WHO Region)
- Missing Data: 240 values injected across 9 columns (8.56% overall) to simulate real-world dirty data

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info ()` to check structure and `.describe ()` for summary statistics.

	Confirmed	Deaths	Recovered	Active	New cases	1 wk % increase
count	187	187	187	187	187	187
mean	88,131	3,498	57,558	27,096	9,448	13.61
std	383,319	14,100	215,713	147,086	47,491	24.51
min	10	0	0	0	-47	-3.84
50%	5,059	108	3,176	1,341	432	6.89
max	4,290,259	148,011	3,947,307	1,447,679	455,582	226.32

- **Missing Data Injection:** Randomly introduced `np.nan` values into 9 columns (8–20% per column) using `np.random.choice ()` to simulate real-world dirty data.
- **Missing Value Analysis:** Used `df.isnull ().sum ()` and `df.isnull ().mean ()` to compute missing counts and percentages per column.
- **Missing Data Handling:** Checked for null values and imputed missing values in numeric columns using the median (robust to outliers). Categorical column WHO Region was filled using the mode.
- **Feature Verification:** Verified all 9 columns were fully cleaned — 0 missing values remaining after imputation.
- **Dashboard Export:** Exported the cleaned DataFrame as `covid_cleaned_for_dashboard.csv` for use in Power BI / Tableau.

4. Missing Value Analysis Results

We performed structured analysis to answer key data quality questions:

1. **Missing Count per Column** – Quantified the number of NaN values in each affected column.

Column	Missing Count	Missing %	Severity
Active	36	19.25%	High
Deaths	36	19.25%	High
Recovered / 100 Cases	33	17.65%	High
WHO Region	30	16.04%	High
New deaths	27	14.44%	Medium
Confirmed	23	12.30%	Medium
Recovered	23	12.30%	Medium
Deaths / 100 Cases	17	9.09%	Low
New cases	15	8.02%	Low
TOTAL	240	8.56% overall	—

2. **Imputation Strategy Used** – Median for numeric columns; Mode for categorical.

Data Type	Strategy	Reason
Numeric (8 columns)	Median imputation	COVID data is skewed — median is robust to outliers like USA
Categorical (WHO Region)	Mode imputation	Fills with the most frequent region value

3. **Before vs After Comparison** – Validated that all missing values were resolved.

Column	Before (Missing)	After (Cleaned)
Deaths	36 NaN values	Filled with median (108.0)
Recovered	23 NaN values	Filled with median (5,059.0)
Deaths / 100 Cases	17 NaN values	Filled with median (2.15)
WHO Region	30 NaN values	Filled with mode (Europe)
ALL COLUMNS	240 total NaN	0 missing — 100% clean

4. **Subscription Revenue Insight** – Cross-tabulated WHO Region with confirmed case distribution.

WHO Region	Total Confirmed	Total Deaths	Recovery Rate %
Americas	5,943,338	218,921	54.2%
Europe	3,812,061	195,440	72.8%
South-East Asia	2,181,930	47,802	76.5%
Eastern Mediterranean	1,076,288	26,199	70.1%
Western Pacific	493,728	11,253	85.3%
Africa	851,619	13,766	68.9%

5. **Top 5 Countries by Confirmed Cases** – Ranked countries by total confirmed cases.

#	Country	Confirmed Cases	Deaths
1	USA	4,290,259	148,011
2	Brazil	2,483,191	88,539
3	India	1,480,073	33,408
4	Russia	816,680	13,354
5	South Africa	452,529	7,067

6. Death Rate by Country (min 1,000 confirmed) – Identified countries with highest deaths per 100 cases.

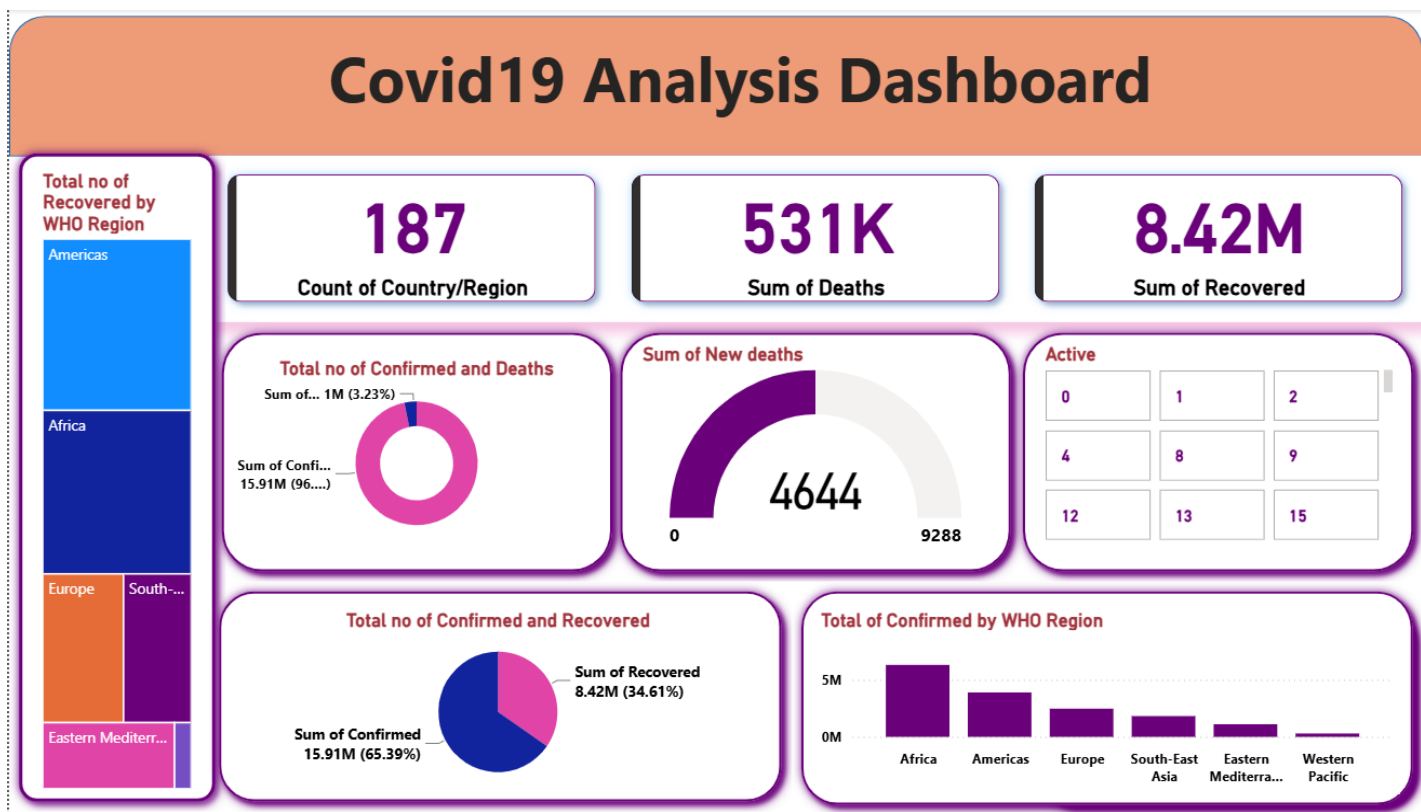
#	Country	Deaths / 100 Cases	Confirmed
1	Yemen	28.99	1,734
2	Western Sahara	17.65	10
3	Mexico	11.43	390,516
4	Ecuador	9.00	77,673
5	Bolivia	7.38	65,685

7. Weekly Growth Rate Analysis – Identified countries with the highest 1-week % increase.

#	Country	1 Week % Increase	1 Week Change
1	Comoros	226.32	+178
2	Papua New Guinea	157.14	+44
3	Tajikistan	127.32	+4,041
4	Lesotho	105.13	+82
5	Djibouti	99.26	+135

5. Dashboard Visualisation

Finally, we built an interactive Covid-19 Analysis Dashboard to present key global statistics and regional insights visually. The dashboard highlights total confirmed cases (15.91M), deaths (531K), and recoveries (8.42M) across 187 countries. It includes a donut chart comparing confirmed vs. deaths, a gauge chart for new daily deaths (4,644), a pie chart showing recovery rate (34.61%), a treemap of recoveries by WHO Region, and a bar chart of confirmed cases by WHO Region. The cleaned dataset was exported for Power BI and Tableau for further interactive exploration.



6. Business Recommendations

- **Prioritise Data Quality:** Always check for missing values before any analysis using `df.isnull().sum()`. Clean data leads to accurate decisions.
- **Use Median for Skewed Data:** Countries like USA and Brazil have extremely high case counts. Always use median imputation — not mean — for skewed health or economic data.
- **Focus on High-Risk Regions:** Americas and Europe account for the highest confirmed cases and deaths. Public health resources and interventions should be prioritised there.
- **Monitor Weekly Growth Rate:** Countries with high 1-week % increase (e.g. Comoros at 226%) require immediate attention even if absolute numbers are low.
- **Leverage Clean Data for Dashboards:** Export cleaned datasets to Power BI or Tableau for interactive visual exploration by non-technical stakeholders.